INSTITUTE OF DISTANCE LEARNING

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI, GHANA

AE 351

Soil and Water Engineering I

Lecture 4

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SURFACE RUNOFF AND HYDROGRAPH ANALYSIS

Estimating Surface Runoff

An important formula for estimating the peak runoff rate is the **Rational Formula**.

It is characterised by:

- consideration of the entire drainage area as a single unit,
- estimation of flow at the most downstream point only and
- the assumption that rainfall is uniformly distributed over the drainage area.
- According to the **rational formula:** $\mathbf{Q_{peak} = CIA/360}$

The Rational Formula follows the assumption that the:

- Predicted peak discharge has the same probability of occurrence (return period) as the used rainfall intensity (I),
- Runoff coefficient (C) is constant during the rain storm, and
- Recession time is equal to the time of rise.
- In the rational formula, it is assumed that discharge starts as soon as rain has replenished the surface retention. Assuming that this occurs at the same time throughout the basin, the order in which the discharge from the various locations in the basin pass through the outlet (Point B) will differ due to differences in their time of travel. The longest time will apply to water travelling from the most hydraulically remote point A. This time is called, $\mathbf{T_c}$ and it is an important hydrological characteristic of the basin.

The Rational Formula

- **Q_{peak} = CIA/360**

- Q_{peak} is peak discharge in m^3/s
- C, the discharge coefficient is basically the proportion of rainfall discharging rapidly as shallow flow. It is a combination of three slope factors i.e. flat, rolling or hilly topography and three categories of land use and three broad soil descriptions. Other variables are ignored.
- I is the intensity of the design storm of duration T_c in mm/h
- A is basin area in ha

C Values

Land Use	C
Business:	
Downtown areas	0.70 - 0.95
Neighbourhood areas	0.50 - 0.70
Residential:	
Single-family areas	0.30 - 0.50
Multi units, detached	0.40 - 0.60
Mufti units, attached	0.60 - 0.75
Suburban	0.25 - 0.40
Industrial:	
Light areas	0.50 - 0.80
Heavy areas	0.60 - 0.90
Parks, cemeteries	0.10 - 0.25
Playgrounds	0.20 - 0.35
Railroad yard areas	0.20 - 0.40

Land Use	C
Lawns:	
Sandy soil, flat, 2%	0.05 - 0.10
Sandy soil, avg., 2-7%	0.10 - 0.15
Sandy soil, steep, 7%	0.15 - 0.20
Heavy soil, flat, 2%	0.13 - 0.17
Heavy soil, avg., 2-7%	0.18 - 0.22
Heavy soil, steep, 7%	0.25 - 0.35
Agricultural land:	
Bare packed soil	
*Smooth	
*Rough	0.30 - 0.60
Cultivated rows	0.20 - 0.50
*Heavy soil, no crop	0.30 - 0.60
*Heavy soil, with crop	0.20 - 0.50
* soil, no crop	0.20 - 0.40
* soil, with crop	0.10 - 0.25
Pasture	0.15 - 0.45
*Heavy soil	0.05 - 0.25
* soil	0.05 - 0.25
Woodlands	
Streets:	
Asphaltic	0.70 - 0.95
Concrete	0.80 - 0.95
Brick	0.70 - 0.85
Unimproved areas	0.10 - 0.30
Drives and walks	0.75 - 0.85
Roofs	0.75 - 0.95

Time of Concentration

- The rational formula assumes that the highest discharge from the basin occurs in response to a storm with a duration equal to T_c because:
- If storm duration is $<T_c$, only part of the basin contributes to the total discharge at the outlet at any one time for the discharge from the area close to the outlet B will have passed through B by the time the discharge from point A arrives
- When storm duration is $>T_c$, there is a time at which the entire basin contributes to the discharge through B although the discharge rates will normally be less than those generated by a storm of duration T_c since rainfall intensities decrease with increasing storm duration

- T_c is determined by the Kirpich Formula:

- $\mathbf{T_c = L^{1.15} / (3080H^{0.38})}$**, hours**

Where,

- T_c is time of concentration in hours
- L is maximum travelling distance in the basin in m
- H is the difference in elevation over the above travelling distance in m

Adjustment of Runoff Coefficient

Table 4.3: Runoff Coefficient Adjustment Factors for Rational Method	
Recurrence Intervals (years)	**C_f**
25	1.1
50	1.2
100	1.25

RIVER DISCHARGE MEASUREMENT TECHNIQUES

- Velocity-Area Method essentially using current meters
- Flow Measurement Structures governed by hydraulic laws using structures like crump weirs, V-notches, flumes, etc.
- Dilution Gauging using chemicals
- Float method where, $\mathbf{V=\frac{0.8L}{t}}$ (where **L** is span of bridge or distance traversed by float, **t** is time taken and **V** is velocity in **m/s**)
- Empirical formulae notably:

Manning's Equation: $\mathbf{V = \frac{R^{2/3}S^{1/2}}{n}}$

Chezy formula: $\mathbf{V = C\,(RS)^{1/2}}$

Note that Discharge, $\mathbf{Q = AV}$

Explanation of Terms and Their Units

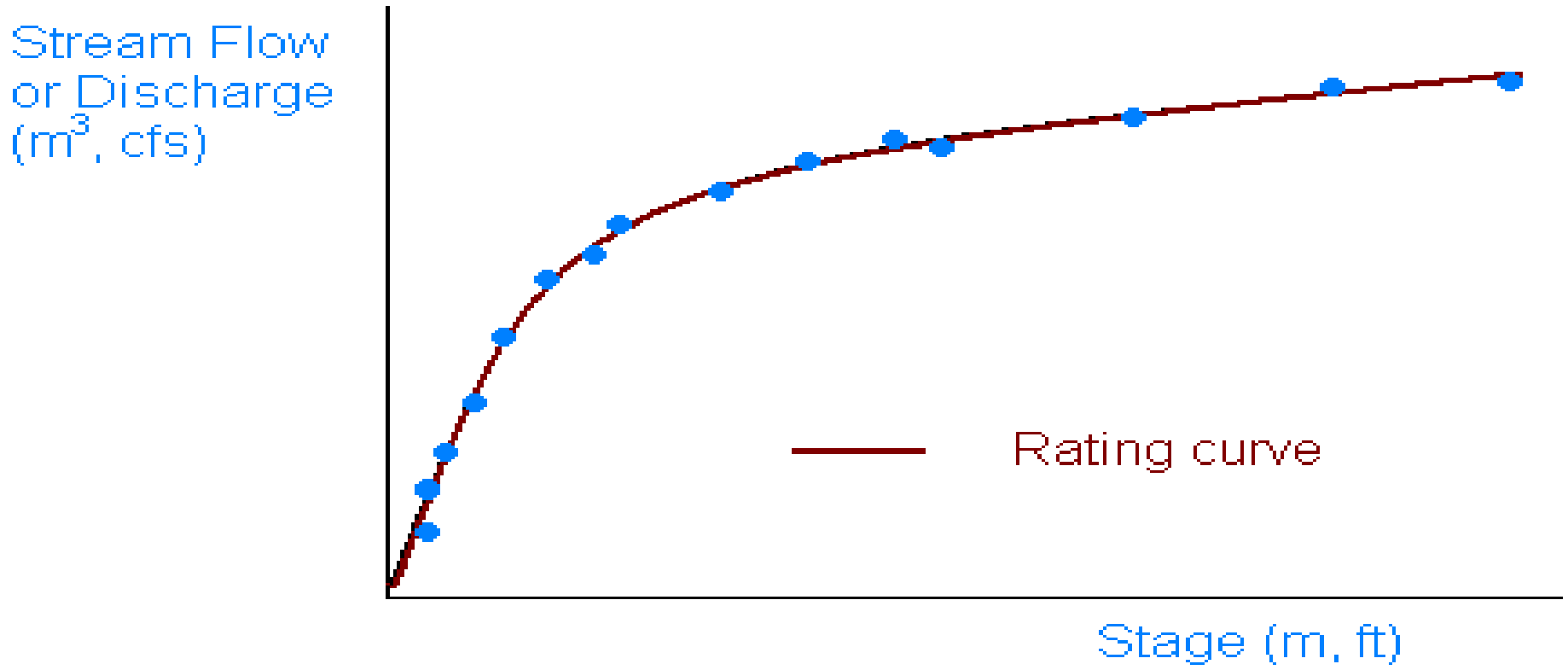
Where,

- V is mean velocity of flow in m/s
- R is hydraulic radius in m (R=Area of flow section in m^2 ÷ Wetted perimeter in m)
- S is slope of the water surface expressed per unit length of the channel or of the energy line
- n is Manning's roughness coefficient which varies with depth of flow, condition of stream bed, condition of stream bank and channel type.
- For natural streams, n is about 0.035
- C is Chezy's constant
- Q is discharge in m^3/s
- A is cross-sectional area in m^2.

STAGE-DISCHARGE RELATION OR RATING CURVE

- A rating curve is a graph that shows the connection between the water level elevation or stage of a river channel at a certain cross-section and the corresponding discharge at the section or
- It is the graphical or numerical relationship between the stage of flow and the corresponding discharge in a stream channel or over a permanent structure. They help to determine the discharge for any stage of flow at any time. They remain constant with time but if it is done for a particular section of a channel, they must be checked from time to time and even after a major flood.
- The stream geometry changes due to scouring and silting of the stream or channel bed. The development of a rating curve involves two steps. In the first step the relationship between stage and discharge is established by measuring the stage and corresponding discharge in the river. And in the second part, stage of river is measured and discharge is calculated by using the relationship established in the first part. Stage is measured by reading a gauge installed in the river.
- If the stage-discharge relationship doesn't change with time, it is called permanent control.

STAGE-DISCHARGE RELATION OR RATING CURVE



- Measurment of stream stage and flow

Use of Current Meters

- **V=a + bN**

Where,

- **V** is velocity
- **N** is in revolutions per minute
- **a** and **b** are coefficients from a current meter obtainable from the manufacturer's specification.

- In current metering, divide the channel into a number of imaginary lines drawn parallel to the stream banks. The velocity at each section is measured at the 0.2^{nd} and 0.8^{th} depth from the surface for medium to large rivers. For rare or shallow situations one measurement at 0.6 depth will approximate to the average over the whole depth

Rating Curve Equation

- For rating curves $\mathbf{Q= k (h-a)^x}$

Where,

- **h** is stage or gauge height
- **k** and x are constants derived from the observed portion of the curve and
- **a** is height I m between zero portion on the gauge and the elevation of zero flow.

- A plot of Q versus (h-a) on semi-log paper is needed to fit a straight line.

Conditions that favour good results for current metering:

- The section should be straight and uniform
- Bed should be smooth and free of vegetal growth, boulders or other obstructions
- Bed and banks should be firm and stable
- The current should be normal to the metering section
- Velocities should be greater than 0.02832 m^3/s and less than 0.11328 m^3/s
- There should be no large overflow section at flow stage and
- The section should be accessible.

A plot of stage versus discharge is alright if parabolic with a dispersion of <2 % about the mean. Larger dispersion implies:

- The control section shifts with scour and deposition in the bed and banks or the growth of vegetation
- That the water surface slope at the control varies as a result of varying backwater from tides, reservoir fluctuation or variable tributary inflow downstream
- That the measurements are not carefully made.

Looping Effect (Further Explanation)

- If a rating curve is plotted by regarding each discharge as being measured on a ***falling*** or ***rising*** stage instead of using the median as is usually done, then we have a looping effect. The variation or looping effect is due to:

- Channel storage: As the surface elevation of a river rises , water is temporarily stored in the river channel
- The variation in surface slope that occurs as a flood wave moves along the channel.

- **NB**: Since the rising stage is associated with the greater slope discharge, measurement taken on a rising stage will plot to the right on the rating curve, and those on the falling stage to the left.

DILUTION GAUGING

- Dilution gauging methods measure stream flow on the basis of the rate of dispersal of an introduced tracer, with the concentration of the tracer monitored downstream. This technique is suitable for small turbulent streams where depths and flows are inappropriate for current metering and flow structures would be unnecessarily expensive and/or permanent.

In this technique the following steps are followed:

- Inject chemical into the stream
- Sampling of the water some distance downstream after complete mixing of the chemical in the water has occurred.
- Chemical addition is by
    - Constant rate injection until a constant concentration is achieved downstream (best method)
    - Gulp injection or single dose (Here the concentration-time correlation is monitored).

Measurements and Tracers Used

- The concentration of chemical is used to compute the dilution and hence the discharge of the stream can be obtained.

- Sample analysis is by automatic colorimetric procedure that estimates the concentration of very small amounts of the chromate or compound used by comparison with a sample of the injection solution. The tracer concentration at the downstream measurement point should rise and stabilize to a constant level.

Tracers used

- Chemical salts, dyes and radioactive materials.
- Chemical salts include, Lithium, potassium, magnesium, chloride and dichromate ions with sodium dichromate as the preferred salt.
- The radioactive materials are titrated water and radioisotopes.

Explanation of Terms

- Tracer concentration in bottle, C_1= Weight of tracer/Weight of deionised water
- Rate of injection = q_1
- Original tracer concentration from upflow = C_O
- If no flow has entered the channel during the mixing of the tracer and assuming no loss of water by seepage and evaporation in the section, the discharge at the downstream, is the discharge Q at the upstream plus q_1 which is the quantity of tracer added.
- C_2 is the concentration of tracer at the downstream.

Derivation of Equation

- $Q\ C_O + q_1\ C_1 = (Q + q_1)\ C_2$
- $Q\ C_O + q_1\ C_1 = Q\ C_2 + q_1 C_2$

Re-arranging

- $Q\ C_O - Q\ C_2 = q_1 C_2 - q_1\ C_1$

Simplifying

- $Q(C_O - C_2) = q1(C_2 - C_1)$
- $Q = q_1(C_2 - C_1)/\ (C_O - C_2)$

- Note that salt concentration is difficult to measure so electrical conductivity which has a linear relationship with concentration is often used.

Another approach for determining flow rate is:

- $Q = \frac{FA}{N}$

Where,

- Q is volume of water flowing in the channel per unit time
- F is counts per unit of radioactivity per unit volume of water per unit of time
- A is total units of radioactivity to be introduced for each discharge measurement
- N is total counts

Chemical salt velocity (NaCl-Time is measured between two conductivity humps):

- $Q = \frac{AL}{t}$

Where,

- A is cross-sectional area
- L is distance between stations
- t is time interval.

STORAGE CAPACITY AND THE MASS DIAGRAM

- Reservoirs are structures that store water and in general, high flows are observed in the rainy season in Ghana
- Regulation of streamflow is required to meet higher demands especially during the dry season. This regulation is possible by constructing reservoirs in the stream.
- When it is desired to use water at uniform or nearly uniform rates greater than the minimum discharge of a river, it is necessary to provide storage capacity in which to impound water during periods of high flows for use during periods of low flows.

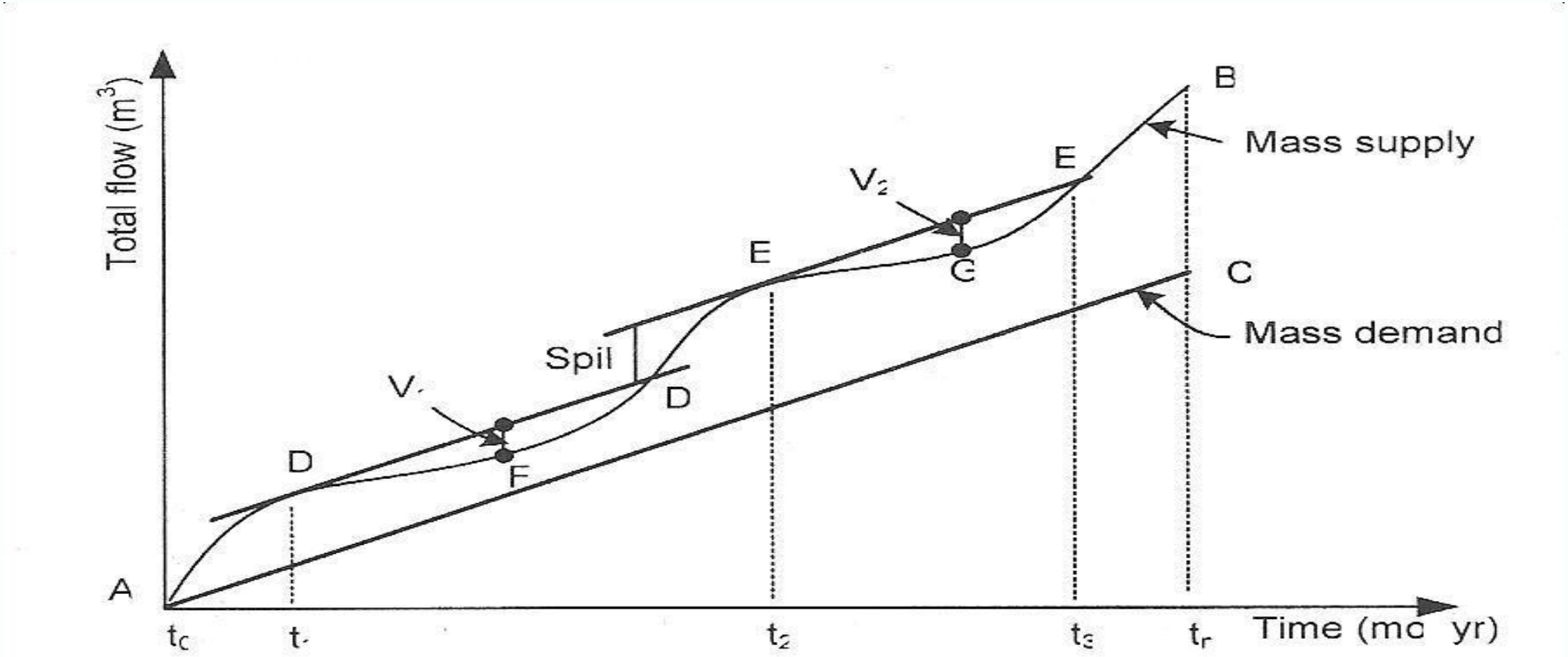
Yield

- The desired yield, *which is the amount of water delivered from a reservoir to meet downstream requirements*, must be available most of the time. For an ideal reservoir site there must be:
- Sufficient impervious and sound formation (bed and side)
- Deep and narrow valley (less evaporation, low cost of expropriation)
- After the determination of location of a reservoir, a special topographic map (1:5000 scale) of the reservoir area is obtained with suitable contour intervals and
- Elevation-Area curve and
- Elevation–Storage curve are obtained.

Mass Diagram

- A mass diagram is used for sizing reservoirs (i.e. minimum size) to provide the minimum average use rate for the stored water.
- A mass diagram in this situation is a graphical representation of accumulated discharge versus time.
- The slope of the mass diagram at any point is equal to the rate of discharge at that time because the difference in the ordinates at the ends of a small segment of such a curve is the volume discharged during the time given by the corresponding difference in the abscissa.
- In other words, the slopes of the diagram are numerically equal to the corresponding ordinates of the hydrograph. Therefore, a mass diagram of a uniform use rate is a straight line.

Determination of Reservoir Yield and Capacity



Explanation of Mass Curve

- The mass curve gives results if ΣD < ΣS during the period of record. The graphical approach is quite satisfactory if the reservoir releases are constant during the period of analysis. When reservoir releases vary, the sequent-peak analysis is recommended.
- From **A** to **D**, the inflow rate exceeds the use rate and the reservoir is full and overflowing
- At **D**, the inflow rate is equal to the use rate and the reservoir is full but not overflowing
- From **D** to **F**, the use rate exceeds the inflow and the amount of drawdown is increasing
- At **F**, the use rate is equal to the inflow rate and the drawdown (**S**) is at a maximum
- From **F** to **D**, the inflow rate exceeds the use rate and the drawdown is decreasing
- At **D**, the reservoir is full and
- From **D** to **E**, conditions are the same as from **A** to **D**

Further Explanation

- Therefore, the greatest vertical distance (**V**) between **DD** and **DFD** occurs at **C**.
-
- **S** is the storage required to maintain the average use rate during the flow period from **D** to **D**.
-
- The largest **V** for the entire record period is the minimum size of the reservoir to provide the minimum average use rate.

HYDROGRAPH ANALYSIS

Basic Definitions

- Base flow sustains rivers during the dry season
- A perennial stream flows throughout the year
- An ephemeral stream flows only occasionally usually during a rainy period
- An intermittent stream flows for part of each year i.e. during or after season of most rain. Such streams could be effluent or influent
- An influent stream is one where the base flow is negative i.e. the stream feeds the groundwater e.g. irrigation canal
- An effluent stream on the other hand, is fed by groundwater and acts as a drainage for bordering aquifers e.g. River Volta and other major rivers in Ghana
- Hydrograph is a graph showing variations of discharge with time, at a particular point of a stream.

Components of Runoff

- Overland flow: That is a fairly uniform amount of water travelling over the surface of the ground until it reaches a channel of any kind. **Horton overland flow** describes the tendency of water to flow horizontally across land surfaces when rainfall has exceeded infiltration capacity and depression storage capacity.
- Interflow: Infiltrated water which move laterally through the upper soil layers until it enters a stream channel
- Groundwater or base flow: Some of the precipitation through deep percolation beyond the root zone of crops might eventually recharge the groundwater. Part of the groundwater which eventually discharges into streams, if the water table intersects the stream channel of the basin, is referred to as base flow which keeps perennial rivers flowing even in the dry season.

- For convenience, total flow is divided into direct or storm runoff and base flow:

- **NB: Total Q – Base flow = Direct Runoff**

Components of a Hydrograph

- The divisions of a hydrograph into direct and groundwater flow as a basis for subsequent analysis is known as hydrograph separation or analysis. Thus, hydrograph analysis provides a way of knowing how much groundwater is flowing from a catchment area upstream of the gauging station in a given time period. For good results, a minimum of
- 5-10 years of data is required.

- If rainfall is unable to infiltrate into any depth, and there is lack of groundwater storage, rainfall events produce a rapid rise in streamflow, which is followed by an almost equally rapid decline at the end of the rainfall.

- For any given year, the volume of water leaving the catchment as base flow must have entered as infiltration. Dividing this volume by the catchment area enables one to express the infiltration in mm to compare with rainfall and evapotranspiration figures.

Summary of Procedure to be Followed

- Plot the total hydrograph from the discharge-time data
- Separate base flow on assumptions that:
  - Direct runoff begins at time zero
  - Direct runoff ceases at point of maximum curvature on recession limb of hydrograph
  - And then draw in the base flow separation as a straight line
  - Or Terminate the direct runoff at a fixed time after the peak of the hydrograph where the time in N days is approximately
- $\mathbf{N=A^{0.2}}$ where **A** is drainage area in km^2
- Calculate discharge values corresponding to direct runoff (i.e. total discharge minus base flow from the hydrograph in Step 2
- Plot duration-discharge data from Step 3 as the runoff hydrograph

Derivation of Unit Hydrograph

- Determine the total net rainfall, i.e. the runoff volume expressed in mm. The net rainfall is now given by the area under the hydrograph (as a volume) divided by the catchment = X
- The unit graph corresponds to 10 mm of rainfall over the catchment, so to derive the unit graph, we must multiply all the ordinates of the runoff hydrograph by 10/X and tabulate the results
- These values in Step 6 are now plotted to derive the unit hydrograph corresponding to the storm of a certain duration
- The hydrograph corresponding to e.g. 6 h storm with total net rainfall of 10 mm can now be constructed by summing 1 h or say 2 h unit graphs (the total of which would correspond to a 6 h storm of 30 mm) and then dividing all ordinates by the number of graphs added in this case 3 (to reduce back to a total rainfall of 10 mm).

FLOOD ROUTING

- A flood is an unusually high stage in a river – normally the level at which the river overflows its banks and inundates the adjoining area.
- The damages caused by floods in terms of loss of life, property and economic loss due to disruption of economic activity are very high.
- Flood peak values are required in the design bridges, culvert waterways, spillways for dams, and estimation of scour at a hydraulic structure.
- At a given location in a stream, flood peaks vary from year to year and their magnitude constitutes a hydrologic series.

Estimation of the Magnitude of a Flood Peak

To estimate the magnitude of a flood peak the following methods are adopted:

- Rational method,
- Empirical equations,
- Flood frequency studies, and
- Unit hydrograph technique.

The use of particular method depends upon the:

- Desired objective,
- Available data and
- Importance of the project.

Definitions and Procedures

- Flood routing may be defined as the procedure whereby the time and magnitude of a flood wave at a point on a stream is determined from the known or assumed data at one or more points upstream.
- Flood routing is the technique of determining the flood hydrograph at a section of a river by utilizing the data of flood flow at one or more upstream sections.
- The hydrologic analysis of problems such as flood forecasting, flood protection, reservoir design and spillway design invariably include flood routing.
- Flood routing shows how a flood wave can be reduced in magnitude and lengthened in time (attenuated) by the use of storage in the reach between two points.
- Generally, routing procedures are based on the relationship between stage and storage. The reduction in peak flows together with the spreading out of the recession curve is referred to as attenuation.

Continuity Equation

- The equation of continuity used in all hydrologic routing as the primary equation states that the difference between the inflow and outflow rate is equal to the rate of change of storage.
- Storage in river reaches:

$$I - Q = \frac{dS}{dt}$$

where I = inflow rate, Q = outflow rate and S = storage. Alternatively, in a small time interval Δt the difference between the total inflow volume and total outflow volume in a reach is equal to the change in storage in that reach

What is needed for routing?

For reservoir routing, the following data have to be known:

1. Storage volume *vs* elevation for the reservoir;
2. Water-surface elevation *vs* outflow and hence storage *vs* outflow discharge;
3. Inflow hydrograph, $I = I(t)$; and
4. Initial values of S, I and Q at time $t = 0$.

Equations

- **$I = D + dS/dt$**

Where,

- **I** is inflow to the reach
- **D or Q** is discharge from the reach
- **t** is the routing period

- **Ds/dt** is the rate of change in reach storage with respect to time

- **$(I_1 + I_2)/2 \times t - (D_1 + D_2)/2 \times t = S_2 - S_1$**

- **$0.5(I_1 + I_2)\, t + (S_1 - 0.5D_1\, t) = (S_2 + 0.5D_2\, t)$**
- The assumption that **$I = (I_1 + I_2)/2$** implies that the hydrograph is a straight line during the period t. Therefore t must be sufficiently short. Routing period t, must never be greater than the time of travel through the reach.
- Routing must be used to compute the hydrograph which will result from a specified pattern of rainfall excess.

Hyetographs

- A hyetograph is the graphical representation of average rainfall and rainfall excess (i.e. rainfall minus infiltration) rates over specified areas during successive unit time intervals during a storm.
- Hyetographs are often plotted above the runoff hydrograph.

Thank You

For any concerns, please contact

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